

# Ultrafiltration

Pre-Lab Plan  
Unit Operations Lab #5  
March 19, 2019



Group #2, Tuesday PM Section  
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## 1. Experimental Objective

The purpose of this experiment is to see how ultrafiltration can separate a concentrate of milk from water. The module includes a feed tank, a permeate milk tank, and a hollow membrane column that consist of fibers outlining the inner diameter of the wall. The permeate flux of the membrane will be determined by pressure drop across the membrane and from milk concentrations.

## 2. Experimental

### 2.1. Apparatus

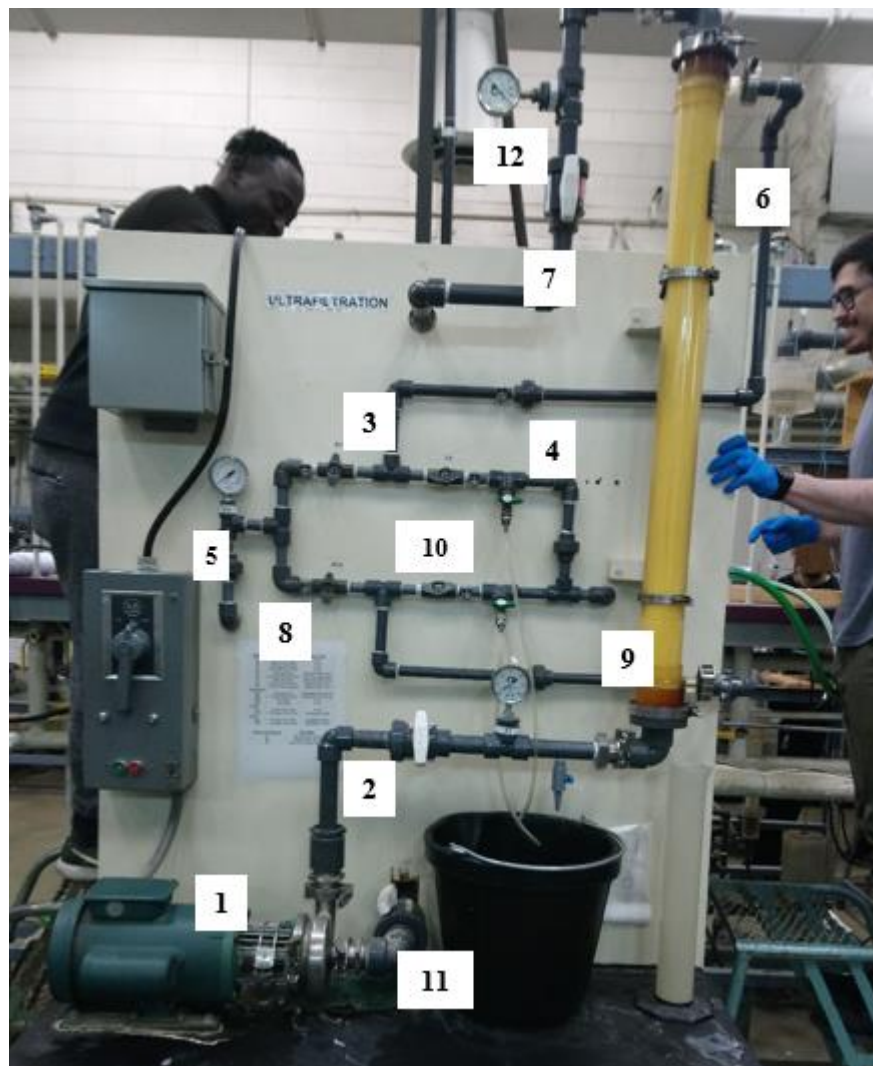
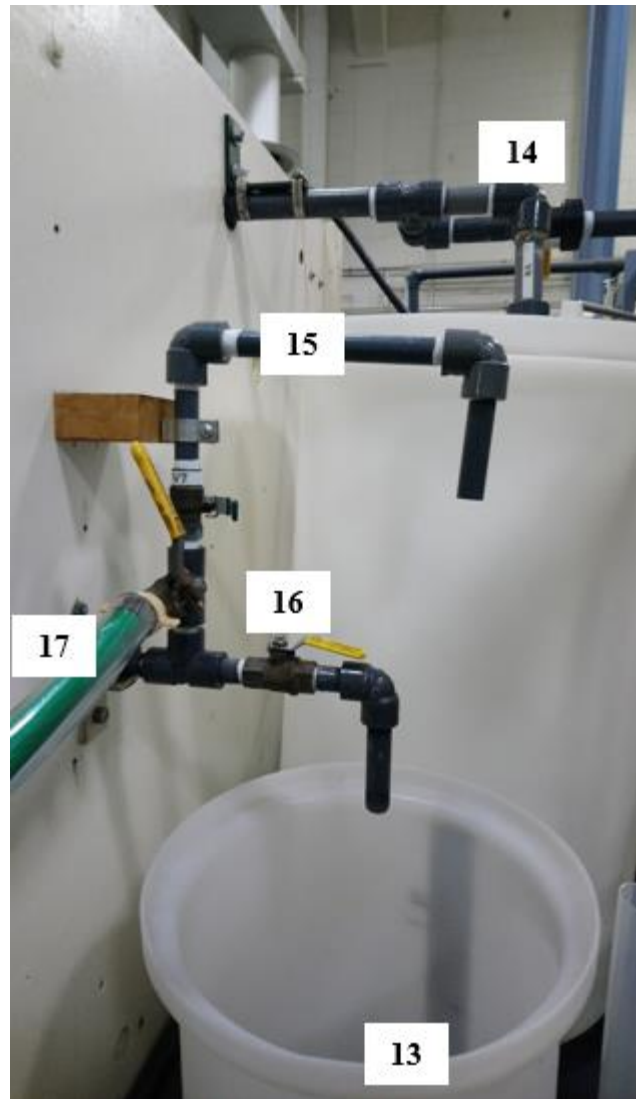


Figure 1: Ultrafiltration Apparatus



*Figure 2: Ultrafiltration Permeate Side*



Figure 3: Ultrafiltration Back Flush Side

Table 1: Apparatus Name and Description

|   | Apparatus | Description                           |
|---|-----------|---------------------------------------|
| 1 | Pump      | Used to Transfer milk solution        |
| 2 | V1        | Inlet Pressure Valve                  |
| 3 | BV3       | Backflush Valve                       |
| 4 | V3        | Permeate Flow Valve                   |
| 5 | P3        | Backflush Pressure Gauge              |
| 6 | Membrane  | Used to filter proteins from solution |

|    |               |                                |
|----|---------------|--------------------------------|
| 7  | V2            | Outlet Pressure Control Valve  |
| 8  | BV4           | Backflush Valve                |
| 9  | P1            | Inlet Pressure Gauge           |
| 10 | V4            | Permeate Flow Valve            |
| 11 | PV1           | Intake-Pump Valve              |
| 12 | P2            | Outlet Pressure Gauge          |
| 13 | Permeate Tank | Used to Collect Permeate       |
| 14 | V8            | Process Tank Return            |
| 15 | V5            | Permeate Drain                 |
| 16 | V7            | Permeate Drain                 |
| 17 | V6            | Permeate Recycle Valve         |
| 18 | Solution Tank | Holds Solution to be separated |
| 19 | BV1           | Recycle Valve                  |
| 20 | V9            | Process Return Bypass          |
| 21 | BV2           | System Return Valve            |
| 22 | Pump          | Moves Solution through system  |

## 2.2 Materials and Supplies

Table 2: Materials and Supplies

|   | Materials        | Description                               |
|---|------------------|-------------------------------------------|
| 1 | Water            | Water used in the system                  |
| 2 | Phosphoric Acid  | Acid used in cleaning process             |
| 3 | Sodium Hydroxide | Caustic solution used in cleaning process |
| 4 | Bleach           | Used in Cleaning Process                  |
| 5 | Stirring Rod     | Used to mix fluids                        |

|    |              |                                        |
|----|--------------|----------------------------------------|
| 6  | Thermometer  | Used to measure temperature            |
| 7  | Litmus Paper | Used to measure the approximate pH     |
| 8  | pH Meter     | Used to measure the pH more accurately |
| 9  | Bucket       | Fluid containment equipment            |
| 10 | Milk         | Fluid that will be separated           |
| 11 | Pipet        | Used to transfer solutions             |
| 12 | Beaker       | Fluid containment equipment            |

## 2.3 Experimental Plan

### *Cleaning Procedure*

- 1.) Fill process tank with 20 L of deionized water at approximately 105°F.
- 2.) Open valve PV1 and direct the permeate outlet flow to the process tank.
- 3.) Close valve V1 and open valve V2.
- 4.) Start the circulation pump.
- 5.) Slowly open valve V1 until a pressure of 20 psi is read on P1.
- 6.) Slowly close valve V2 until a pressure of 8-10 psi is read on P2.
- 7.) Readjust V1 until inlet pressure is stabilized at 25 psi and the outlet pressure is 10 psi.
- 8.) Allow system to run for 10-15 minutes and check the system for leaks. (There should be no air bubbles.)
- 9.) Slowly open valve V2 fully and close valve V1.
- 10.) Stop the circulation pump and drain the water from the process tank and filter.
- 11.) Rinse the process tank.
- 12.) Make up 20 L of approximately pH 2.00 phosphoric acid solution in deionized water at 105°F.
- 13.) Repeat steps 2-7 and run the process for 10 minutes.
- 14.) Close off valves V3 and V4 and run acid solution through the filter for 5 minutes.
- 15.) Repeat steps 9-11.
- 16.) Refill the process tank with 50 liters of 105°F deionized water.
- 17.) Direct the permeate outlet flow to the permeate tank.
- 18.) Open valve DV2 and repeat steps 3-7.
- 19.) Momentarily crack the drain valves during rinse cycle to flush out trapped acid solution.

- 20.) Make sure to stop pump before water level in process tank falls below the level of the inlet pipe to the pump.
- 21.) Rinse the permeate tank.
- 22.) Make up 20 L of a 1% sodium hydroxide solution in deionized water at 105°F.
- 23.) Repeat steps 2-11 but direct the permeate to the permeate tank instead of the process tank. (Make sure drain valve of permeate tank is closed, permeate solution is to be saved.)
- 24.) Close valve V6 and open PV2.
- 25.) Close valve DV2 and open V5.
- 26.) In the back flush area, close valve BV1 and open BV2.
- 27.) In the front, open BV4, BV3, and V2.
- 28.) All other valves in the front should be close except PV1.
- 29.) Turn on pump C and allow permeate liquid to flow into the process tank.
- 30.) When permeate level halves, close DV1.
- 31.) Drain permeate tank, close all back flush valves, and open valve F.
- 32.) Turn off back flush pump and turn on pump G.
- 33.) Slowly open inlet pressure valve to 20 psi.
- 34.) Slowly close outlet pressure control valve to 10 psi.
- 35.) Adjust valves until the inlet is set to 25 psi and the outlet is set to 10 psi.
- 36.) Run the system for 15 minutes and check for leaks in the system or air bubbles in the tank.
- 37.) Slowly open the outlet pressure valve and close the inlet pressure valve.
- 38.) Drain the filter and the process tank.
- 39.) Rinse the process tank.
- 40.) Make up a 200 ppm sodium hypochlorite solution in 20 L of deionized water at 105°F. (use 350 mL household bleach/50 L water)
- 41.) Repeat steps 2-11 and 16-21.

#### *Water Flux Test*

- 1.) Determine the permeate water flux of the ultrafilter at an inlet pressure of 25 psi and outlet pressures of 5, 10, 15, and 20 psi using 20 L of 105°F deionized water.
- 2.) Flux should be measured in a graduated cylinder with a timelapse of at least 30 seconds.

#### *Milk Test*

- 1.) Fill the process tank with 80 L of 105°F deionized water.
- 2.) Add 1 gallon of milk and stir **(do not start the system until the feed material is at the correct temperature)**
- 3.) Start the system and adjust the pressure to 25 psi at the inlet and 5 psi at the outlet.
- 4.) Run in this manner until 10% recovery is reached.
- 5.) Return the permeate line to the process tank and maintain this operation mode(no recovery) while performing the permeate flux measurements.

- 6.) Flow rate is changed by increasing or decreasing V2.
- 7.) When changing from one pressure to the next, wait two minutes before taking a reading.
- 8.) It is recommended that the process outlet pressure is varied in the following order: 5, 15, 10, 20, and 5 psi.
- 9.) Repeat the experiments at recoveries of 25% and 40%.

|                              |                                                               |
|------------------------------|---------------------------------------------------------------|
| <b>Independent Variables</b> | Pressure<br>Temperature<br>Recoveries<br>Concentration Factor |
| <b>Dependent Variable</b>    | Permeate Flux<br>Concentration                                |

### Project Safety Evaluation

Project Title: Ultrafiltration  
Date: 19 March 2019

|                                | Yes/No |                                                                                                                                |
|--------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------|
| Potential electrical hazards?  | Yes    | <i>If yes, identify conditions:</i><br>Pump that is used in experiment must not come in contact with water to prevent hazards. |
| Potential mechanical hazards?  | Yes    | <i>If yes, identify conditions:</i><br>Pump needed in experiment.                                                              |
| Potential pressure hazards?    | Yes    | <i>If yes, identify conditions:</i><br>High pressure being sent to the membrane                                                |
| Potential temperature hazards? | No     | <i>If yes, identify conditions:</i>                                                                                            |



|                                                   |                     |                                                                                                                     |
|---------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Hazardous/toxic chemicals involved?               | Yes                 | <i>If yes, hazards involved:</i><br>Sodium Hydroxide<br>Phosphoric Acid<br>Sodium Hypochlorite                      |
| Gloves required?                                  | Yes                 | <i>If yes, specify type:</i><br>Chemical resistant gloves while working with bases and acids                        |
| MSDS reviewed by all team members?                | Yes                 | Sodium Hydroxide      CAS 1310-73-2<br>Phosphoric Acid      CAS 1310-73-2<br>Sodium Hypochlorite      CAS 7681-52-9 |
| <i>Process Parameters</i>                         | <i>Max Expected</i> | <i>List location and possible hazards and precautions</i>                                                           |
| Temperature (°C)                                  | 40°C                | Water to tank temperature                                                                                           |
| Pressure (psia)                                   | 25                  | Feed to membrane pressure                                                                                           |
| <i>Safety Controls</i>                            | <i>yes/no</i>       | <i>If yes, Provide description</i>                                                                                  |
| Safety Switch<br>Pressure valves<br>Thermocouples | Yes                 | Used to turn on/off pump<br>Control pressure<br>Control temperature                                                 |

I have read relevant background material for the Unit Operations Laboratory entitled: “Ultrafiltration” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Chris Kozlowski      3/19/2019  
Signature of team member      Date

Aref Wahdan      3/19/2019  
Signature of team member      Date

Anna Borkowska      3/19/2019  
Signature of team member      Date

Alexander Kubinski      3/19/2019  
Signature of team member      Date

Eric Adupong

3/19/2019

Signature of team member

Date